



nal components, thus minimising the engineering bill of materials (eBOM). Its design operates on a system clock with an on-chip RCX, eliminating the need for a sleep mode crystal in most applications. The SoC's packaging options reduce the solution footprint. *Renesas Electronics Corporation*
<https://www.renesas.com>

World's first Wi-Fi Certified 7 tri-band single-chip solution

MaxLinear has launched the world's first Wi-Fi Certified 7 tri-band single-chip solution. The Wi-Fi access point enhances speed, capacity, and efficiency, making it well-suited for



areas with high demand. It features cutting-edge technologies such as 4K QAM, multi-link operations, multi-resource unit, and puncturing, multi-user multiple input multiple output, orthogonal frequency division multiple access, advanced beamforming, and improvements in power efficiency. The product achieves a peak throughput of 11.5Gbps on the 6GHz (6E) band, offering speeds that are double those of the previous Wi-Fi 6 generation.

MaxLinear
<https://www.maxlinear.com>

Industry-first 28nm radar one-chip family

NXP Semiconductors has announced an addition to its automotive radar chip family with the SAF86xx. Along with



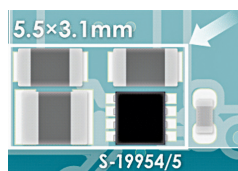
NXP's S32 processors, vehicle connectivity, and power management, this system solution supports the development of software-defined radar tech-

nologies. The radar SoC streams radar sensor data at up to 1Gbit/s, enabling automakers to refine ADAS partitioning for software-defined vehicles and transition to new architectures. It allows original equipment manufacturers to update and add radar functionalities throughout the vehicle's lifespan via over-the-air updates. The product uses 28nm RFCMOS technology to enhance radar sensor performance beyond older 40nm or 45nm technologies.

NXP Semiconductors
<https://www.nxp.com>

Industry's smallest package switching regulators

ABLIC has launched synchronous step-down switching regulator S-19954/19955 series, which the company claims to be the industry's smallest package switching regulators. The IC, developed through CMOS process



technologies, is a secondary step-down switching regulator with an integrated Power Good function. It allows the choice between PWM control or PWM/PFM switching control. The S-19955 series, with PWM/PFM switching control, uses PWM control for heavy loads and switches to PFM control for light loads, providing operation based on the load condition. With a switching frequency of 2.25MHz, the IC supports the use of smaller external components, making the IC suitable for applications where space is a concern.

ABLIC
<https://www.ablic.com>

Neuromorphic SoC that mimics brain's data processing

Innatera has launched a neuromorphic microcontroller, the Spiking Neural Processor T1. The processor uses neuromorphic processing technology

that mimics the brain's mechanisms for sensory data processing. The T1 system-on-chip aims to transform sensor data processing at the edge, enabling applications in wearables, smart homes,

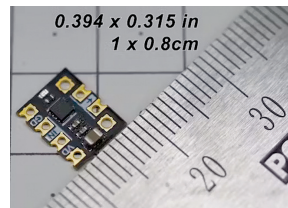


and IoT devices. It offers event-driven processing capabilities for sensing scenarios. At its core, T1 integrates neuromorphic computing technology, a RISC-V processor, and support for CNN models. The processor is supported by a software development kit, Talamo. Integrated with the PyTorch framework, Talamo provides a platform for developing and deploying spiking neural network applications, serving as a tool for developers interested in neuromorphic processing.

Innatera
<https://innatera.com>

World's smallest high-current driver module

Microbots have launched DriveCell, a compact h-bridge module, measuring 0.8cm x 1.0cm, designed for controlling high current loads, such as brushed motors, LED strips, or other Microbots



devices, including CoilPad and FlatFlap. It can be operated through an Arduino using a library supplied by Microbots. It is equipped with the DRV8837 chip, designed for precise control of output current direction through its two input pins, which are typically activated in opposite polarities. The chip comes with several safety features, including overcurrent protection, undervoltage lockout, and thermal shutdown, necessitating caution to avoid reversing its supply rails.

Microbots
<https://microbots.io>